dan cascaval

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Education

Carnegie Mellon University *Pittsburgh, PA*

B.A. • Expected Majors: Computer Science, Architecture • GPA 3.76 *08/15 - 05/20*

Select Courses Compiler Design, Parallel Computer Architecture/Programming, Software Design/Concurrency Parallel Algorithms/Data Structures, Functional Programming, Generative Modeling, Calculus 3D

Distinction Dean's List, 5/6 semesters; SoA School Honors, *All Studios*; Student Senate Committee Vice Chair

Experience

Digital Practice Intern, NBBJ *06/18 - 08/18*

- Designed and developed design computation tools, interviewing, iterating and testing directly with end-users.
- Architected and wrote plugins (C#/Rhino3D) merging geometry with data and propagating design changes.
- Conceptualised and implemented constraint solving for extensible library of reusable parametric objects.

Research Assistant, Manufacturing Futures Initiative *06/17 - 12 /17*

- Procedurally generated complex fabricable surfaces with physical simulation and evolutionary algorithms.
- Scripted toolpathing and code for 3D-printed robotic fabrication of arbitrary panels using 6-Joint robot arms.
- Developed system for visualising and iterating on design in response to surface-actuated thermal changes.

Teaching Assistant, Carnegie Mellon University

Head TA, Digital Media I/II *08/16 - 05/18*

- Create and deliver workshops on 3D modeling, scripting, rendering, animation, and digital design.
- Design, develop, and execute 2-week projects for multi-semester course in computational media.

TA, Principles of Software Construction: Objects, Design, and Concurrency *08/18 - pres.*

- Lead recitation, provide design feedback on student work, grade student coursework.

TA, Interbreeding Architecture (Computational Techniques for Built Environment) *01/18 - 05/18*

- Create 3D training workshops, environmental analyses, give feedback on design research projects.

Projects

Impala : Parallel Component Framework for Grasshopper *Independent, 2017 - 18*

- Implemented and refined parallel algorithms to perform 100,000+ object in-memory physical simulations.
- Devised extensible, object-oriented, method generation patterns to maximise parallelism and performance.
- Improved baseline runtime on over 20x in practice on several common patterns via parallelism and tuning.

GeoFilter : Spatial Data Processing Framework *Course Project, 2018*

- Architected and implemented framework including data input, analysis, visualisation plugins and GUI graph.
- Won Best Framework (of ~40 projects) and created documentation and support for extending functionality.

Skills

Languages/Libraries C#, Java, Python, C/C++, SML/OCaml, Git, PThreads, JavaScript, Processing, OpenGL, G (CNC)

Software Grasshopper3D, Rhino3D, Adobe Creative Suite, AutoCAD/Inventor, Houdini, Autodesk Revit

Hardware 3D Printing (PLA/ABS/Resin/Powder), 6-Axis Robotic Arm, Laser cutting, CNC Milling/Routing

Design Software (Object Oriented, Functional), Web (UI/UX, JS), Architectural (Buildings)